Toronto Transit Commission Bus Delay Analysis and Prediction

Raihanah Nezar AlAli
CS Department
King Faisal University
Al-Ahsa, Kingdom of Saudi Arabia
ID: 223007420

Arwa Ahmed Alabdulatif
CS Department
King Faisal University
Al-Ahsa, Kingdom of Saudi Arabia
ID: 223041726

Yasmeen Abdullah Al-Besher
CS Department
King Faisal University
Al-Ahsa, Kingdom of Saudi Arabia
ID: 223035718

Dina Mohammed Aljughayman
CS Department
King Faisal University
Al-Ahsa, Kingdom of Saudi Arabia
ID: 223024181

***Abstract*—Public transportation delays can cause unreliable services and issues for those who depend on public transport for daily activities. This project examines historical Toronto Transit Commission (TTC) bus delay data to identify patterns of delays and support future delay prediction using machine learning. The dataset includes data from 2021 to 2026 about bus routes, incident type, location, and delay duration. The dataset was preprocessed through integration of data from various sources, standardization of column names and formats, filling of missing values, and removal of invalid and duplicated records. After the preprocessing, there were 317,758 records and 11 columns in the dataset. Exploratory data analysis (EDA) and visualization were done to understand the distribution of delays and to identify the factors affecting delay length. It was found that most delays were relatively short, but some incidents caused very long delays. Differences in delay durations were observed based on incident type, route, and time periods. The cleaned dataset and EDA results provide a foundation for developing a Random Forest machine learning model to predict bus delay durations.**

Keywords*— *public transportation, bus delay analysis, exploratory data analysis (EDA), predictive model

I. INTRODUCTION

Public transportation is a key component of mobility in cities. If it is regular and timely, people do not have problems getting to work, school, or meetings. In Toronto, Canada, the city's subway, streetcar, and bus services are run by the Toronto Transit Commission (TTC), which serves as one of the main ways residents move around the city.

To support transparency, the City of Toronto shares the TTC bus delay history through its open data portal, detailing for each incident the date and time, the route and location, a delay code, and the length of the delay in minutes [2]. These public records offer a valuable opportunity to better understand how and why service disruptions occur across the bus network.

A. Problem Statement

Public transit delays can make travel longer and lower the reliability of service, on TTC network. TTC shares delay records, but the raw data often has missing values, inconsistencies and other quality problems that hurt the analysis. Also, looking at public transit delays at several times and locations is hard. Without data careful exploration and good prediction models, it becomes hard to see when and where public transit delays happen to understand the causes and to forecast future public transit delays from historical data.

B. Objectives

The primary objective of this project is to clean and preprocess the historical TTC delay dataset by handling missing values and inconsistencies to improve data quality. Furthermore, the study aims to conduct exploratory data analysis (EDA) to identify temporal and spatial delay patterns and investigate factors associated with reported delays. Finally, the project focuses on developing and evaluating a machine learning model to predict transit delay durations based on historical data and available incident features.

II. LITERATURE REVIEW

This section analyzes five previous studies that have been done in the field of transit delays analysis and prediction. In addition, important aspects of each study, such as dataset selection and preprocessing, machine learning models, and final results are discussed in this section to identify the main techniques that may support our project. The following resources have been identified:

A. Interpretable Machine Learning for Identifying the Determinants of Bus Delays

This paper was published in March 2026 [1]. It discusses the machine learning approach the researchers proposed to predict bus delays. The dataset used contains on-board motion recorders that continuously record vehicle position, speed, time, and other real-time parameters that might be relevant to the bus delays. Researchers have selected only one bus route and used gradient-boosted decision trees as predictive models due to their ability to perform well with lower computational costs. The models used to predict the bus delays are Random Forest (RF), Extreme Gradient Boosting (XGBoost), and Categorical Boosting (CatBoost). They showed close prediction performance when evaluated by three metrics, which are Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R^2 (Coefficient of Determination). RF slightly outperformed the other models, achieving an RMSE of 5.369 minutes, an MAE of 3.633 minutes, and an R2 of 0.445. The most influential factors identified were time of day, stop sequence, and weekdays [1].

B. Measurement and Classification of Transit Delays Using GTFS-RT Data

This paper was published in 2022 [3]. It introduces an approach for calculating bus delay times using GTFS-RT, the real-time part of the General Transit Feed Specification, which is a standardized public format that can easily be used to share bus positions. The researchers collected GTFS-RT bus data for one whole year from King County Metro in Seattle, Washington. After cleaning, they ended up with about 27 million tracked bus locations. Delays were separated into two types: systematic delays, which refer to predictable delays such as traffic congestion, traffic signals, and bus stop time, and stochastic delays, which refer to delays caused by unexpected events, for example, collisions or a high volume of passengers at a specific point. Every bus delay was assigned to its corresponding street segment, which made it possible to locate where delays occur most across the network. The median systematic delay was 95 seconds, while the median stochastic delay was -19 seconds. This implies that buses often ran ahead of schedule, possibly due to excessive schedule padding. The slowest segments were identified in the central business districts [3].

C. Unsupervised Learning for Public Transport Delay Pattern Analysis

This paper was published in 2024 [4]. It shows how unsupervised learning can be applied to identify delay patterns on public transport at the single-line level. Using Automatic Vehicle Location (AVL) and General Transit Feed Specification (GTFS) data, the authors conducted a case study of tram line 1 in The Hague, Netherlands, covering a period of 79 days during the summer of 2019. To avoid contamination of the data caused by rare events such as equipment failures, they kept only the records with valid stop names and removed extreme values. The cleaned data was then converted into images showing the average hourly delay at each stop for every day, which can be considered visual summaries of each day's operation. These images were processed by a pre-trained neural network (ResNet50) to extract their features and then grouped using the k-means algorithm, which revealed five distinct daily

delay patterns. The results showed that delays were usually more severe on weekends than on weekdays, with trams being most punctual on Mondays and least punctual on Sundays [4].

D. Public Transport Delay Prediction using Deep Learning

This paper appeared in 2026 [5]. Offers a deep learning way to forecast public transport delay prediction using travel and weather data. I think the authors approach to public transport delay prediction is compelling. The dataset contains streetcar delay records from the TTC and daily weather information in Toronto from 2014 to 2019. The authors compared machine learning and deep learning models, such as Support Vector Machine (SVM), Random Forest, Long Short-Term Memory (LSTM), and Deep Neural Network (DNN). The findings show that DNN delivered the results with a Mean Absolute Error (MAE) of 3.21 minutes for public transport delay prediction compared to 3.37 minutes for LSTM and 3.60 minutes, for SVM [5].

E. Real-time Bus Arrival Delays Analysis Using Seemingly Unrelated Regression Model

This paper was published online in June 2024 [6]. The research focused on investigating the factors influencing bus arrival delays using real-time bus operation and weather data from Stockholm, Sweden. The researchers used data from bus line 4 collected between January and June 2022. After data preprocessing, the dataset included 20,189 bus records in one direction and 20,108 records in the opposite direction. The authors applied the Seemingly Unrelated Regression Equations (SURE) model to identify the effects of different factors on delays at individual bus stops. The results showed that previous stop delays, dwell time, scheduled travel time, and traffic conditions had a greater influence on bus arrival delays compared to weather and calendar variables. The model achieved R^2 values ranging from 0.92 to 0.996, indicating that it explained a large proportion of the variation in bus arrival delays. However, the research focused on a single bus route, which may limit the generalizability of its findings. This study is relevant to our project because it identifies factors associated with bus delays, while our project focuses on analyzing historical TTC bus delay records and developing a Random Forest model to predict delay durations.

F. Dataset and Methodology

The dataset selected is published by TTC, containing the historical bus delay records since 2014 [2]. This study will use records from 2021 to 2026 to analyze recent delay patterns and potential changes after the COVID-19 pandemic. The dataset contains temporal information, route, location details, incident types, and delay durations. Having several types of features may enhance the prediction process by considering several aspects that might be relevant to the bus delays. However, it may contain missing values that could affect the data analysis accuracy.

The project includes several stages to achieve its goal. First, data cleaning is applied by removing duplicates, fixing data inconsistencies, and handling missing data. Second, data exploration is performed to identify patterns, outliers, and potential features that will be selected as most relevant features to the bus delays. Third, feature engineering extracts useful features from the dataset. Any relevant categorical feature is then encoded into its numerical representation. Normalization is not required because the Random Forest (RF) model that will be used is not sensitive to differences in feature scales. Finally, the prepared data is divided into training and testing sets and then supplied to train and evaluate the RF model. The RF model will

combine multiple decision trees to identify relationships between the features that may cause delays.

This project is implemented in Python using Google Colab and Jupyter Notebook, with the following libraries:

1. **Pandas:** Used to load, integrate, and clean the TTC bus delay records and to summarize delays by route, incident type, day, and time period.
2. **NumPy:** Used for numerical operations, such as fitting a trend line between delay duration and the gap between buses.
3. **Matplotlib:** Used to create histograms, bar charts, scatter plots, and heatmaps to visualize delay patterns.
4. **Re:** A built-in Python module used to standardize route numbers and vehicle identifiers.
5. **Datetime:** A built-in Python module used to extract the hour of each incident and group delays into time periods.
6. **Scikit-learn:** Used to implement the Random Forest model.

III. Preliminary Analysis

This section presents the results of dataset preprocessing, exploratory data analysis, and data visualization. These steps collectively improve the data quality and provide a better understanding of the dataset before developing the machine learning model.

A. Dataset Preprocessing

The TTC bus delay dataset was preprocessed to enhance the quality and consistency of the data before exploratory data analysis and prediction model development. The dataset is made up of data records from 2021 to 2026 and is extracted from several Excel and CSV files. Initially, the datasets were combined to form one dataset consisting of 319,083 records with 11 columns.

The data standardization process involved the renaming of columns such as Line, Station, Bound, and Code to Route, Location, Direction, and Incident, respectively. Standardization of route and vehicle identifiers was done, and incident codes were converted into categories. Location and incident description names were also formatted.

The presence of missing values was assessed in the dataset. The Direction column had 58,182 missing values (18.23%), and the Route column had 3,057 missing values (0.96%). Missing values were imputed with “Unknown” to retain records for further analysis. In addition, 71 records with invalid direction values were deleted.

Duplicate records were detected and deleted to avoid the influence of repetitive records on the analysis. 1,254 duplicate records were removed. The dataset was then validated to ensure that no duplicate records remained and that missing values in the selected important columns had been handled.

After data preprocessing , the final dataset had 317,758 rows and 11 columns. The cleaned data was saved as a CSV file for further analysis and visualization.

B. Dataset Exploration

Exploratory data analysis was conducted on the cleaned TTC bus delay dataset, which contains 317,758 records across 11 features. The summary statistics for the numerical attributes (Min Delay and Min Gap) revealed an overall average delay time of approximately 20.55 minutes with a standard deviation of 50.83 minutes. Initial analysis indicated significant skewness caused by extreme outliers, with recorded delay durations reaching up to 999 minutes. Utilizing the

Interquartile Range (IQR) method, 16,876 potential outliers were identified in the delay duration variable. Further examination confirmed that extreme delay values exceeding 360 minutes accounted for 1,740 entries, primarily associated with major service diversions and system disruptions. Categorical frequency distributions demonstrated that operational issues, mechanical failures, and route-specific congestion represent the primary drivers behind transit disruptions across the network.

C. Dataset Visualization

The data are presented in several graphs to provide a clear understanding of the dataset. Figure 1 demonstrates the distribution of delay times in minutes; it shows that delays ranging from 0 to 50 minutes are the most frequent in the dataset. Meanwhile, Figure 2 shows that `Parades/Marches` and `Diversion` incidents result in the highest average delays. It illustrates the effect of incidents on average delays.

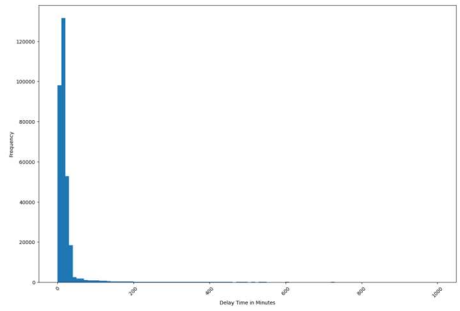


Figure 1: Delay duration distribution.

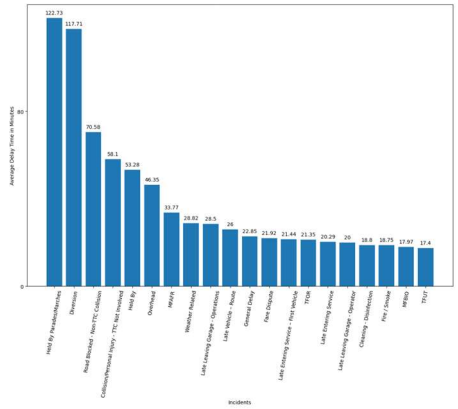


Figure 2:Average delay by incident.

Figure 3 presents the routes ordered by the median delay time, showing that routes may affect the median delay time, The results suggest that a higher median delay does not necessarily correspond to a higher incident count. For example, the `Line Shuttle` route has the highest median delay time, but it has only one incident, meaning there is only one record for that route.

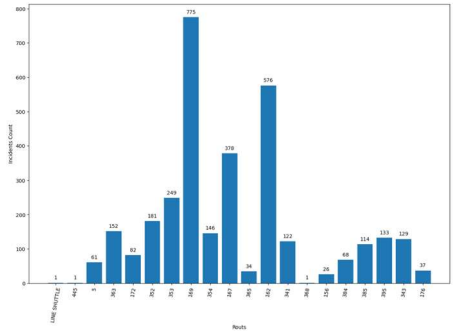


Figure 3: Incident count by route for the 20 routes with the highest median delays.

Figure 4 distinguishes between the median delay time in each time period of the day, showing a high increase in the median delay time at night and in late morning. Finally, the scatter chart shows a strong correlation between the delay time and the time gap between buses, as shown in Figure 5. This indicates that longer delays are associated with longer time gaps between buses at each station. However, it cannot be confirmed whether the time gap between buses is recorded before or after the occurrence of a delay. Therefore, it may be excluded from the model features to reduce the risk of data leakage.

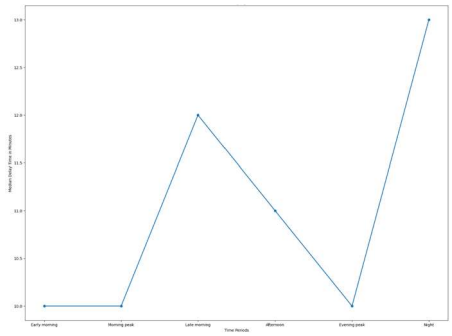


Figure 4: Median bus delay by time period.

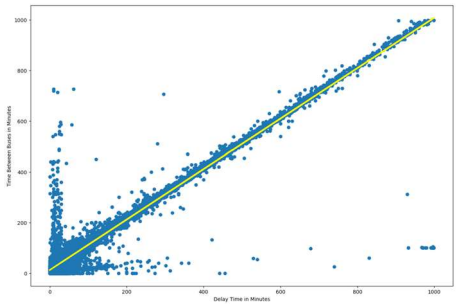


Figure 5: The relationship between delay times in minutes and time between buses.

IV. Conclusion

This project aims to analyze bus delays, identify patterns and features associated with bus delays, and train a machine learning model for prediction. Related research papers have been reviewed to identify their research gaps that could help us in building the project idea. TTC bus delay history from 2021 to 2026 was selected and preprocessed to improve the data quality and handle data inconsistency. Exploratory data analysis and data visualization are conducted to identify important patterns and trends, providing a strong foundation for random forest model training in the next milestone.

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